AV Skills IV: Live Inputs + Tracking

Refer to the Corresponding Max/MSP Patches for Examples of Implementation

0. USEFUL MAX PACKAGES

Package	Description
cv.jit	Computer vision tools (image segmentation, gesture recognition, motion tracking, etc. as well as educational tools that outline basics of computer vision techniques)
Vision	Computer vision tools including body, face, and hand trackers, object detection, depth detection and more.
Vsynth	Virtual Video Synthesis Toolkit
Vizzie / Beap	Prebuilt max video and audio tools
AV Toolbox	Real-time A/V toolkit for signal analysis, timeline control, extended jitter functionality, parameter mapping
VST Audio Plugins	You can search for free VSTs online - just download and install them according to instructions

1. VIDEO INPUTS

Acquire visual information

Video	Purpose	Max Object
Live camera	Webcam, or other external camera	jit.grab
Pre-recorded video	Play video files	jit.movie
Kinect / depth camera	Depth maps (for point clouds)	jit.freenect.grab (external)
NDI / Spout / Syphon	Stream a video from another location into max	jit.ndi.receive (Universal) jit.gl.spoutreceiver (Windows) jit.gl.syphonreceiver (Mac)

2. IMAGE PRE-PROCESSING

Prepare and modify incoming image data

Technique	Purpose	Max Objects
Downsample	**Data downsampling:** Reduce the resolution so there's less data to process, (makes everything faster and helps reduce out noise)	cv.jit.resize
Mirror	Flip the image horizontally so movement matches what the participant sees on screen like a mirror.	jit.dimmap
Grayscale	**Data downsampling:** Convert a colour image to grayscale by collapsing a 4-channel colour image into a single grayscale luminance plane, so there's about a quarter of the data.	jit.rgb2luma
Blur	**Noise Reduction:** Soften the image to remove speckled noise, which stops thresholding from picking up stray pixels.	jit.fastblur cv.jit.blur
Threshold	**Noise Reduction:** Turn a grayscale image into pure black and white. Pixels above a set value become white and the rest become black. Good for filtering out unwanted details of a live feed.	jit.op @op > cv.jit.threshold
Trail	**Temporal Effect:** Keep a fading record of earlier frames so movement lingers and decays gradually, which gives subtler, more gesture-like effects.	jit.slide cv.jit.ravg

3. DETECTION / TRACKING TECHNIQUES

Identify objects, people and movement

Technique	Description	Max Objects
Frame difference	Simple motion detection	jit.op @op absdiff jit.absdiff
Blob Tracking	Track blobs with bounding boxes	cv.jit.blobs.bounds
Centroid Tracking	Follow the center of a tracked object	cv.jit.centroids
Colour Tracking	Track a specific colour	This tutorial Another tutorial
Face Tracking	**Advanced:** Track one or more faces	cv.jit.faces
Body Tracking	**Advanced:** Track body, hands, face, and other more complicated things	Vision Max External
Optical Flow	Detect the direction and velocity of movement across an image	cv.jit
Background Subtraction	Separate moving foreground objects from a relatively static background.	cv.jit
Depth Tracking	Track objects based on their distance from the camera rather than their colour or brightness.	jit.freenect.grab

4. ANALYSIS

Extracts measurable properties of an image.

Calculation	Description	Max Objects
Average luminance	Measure the overall brightness of an image	jit.rgb2luma jit.3m (Use the mean)
Dominant Colour	Detect the dominant colour present in an image.	This Tutorial This Forum Thread jit.histogram
Colour Distribution	Measure how frequently different colours or intensity values occur to identify colour palettes and tone distribution.	jit.histogram
Brightness Distribution	Measure the proportion of pixels that are dark, midtone, or bright.	jit.rgb2luma jit.histogram
Hue / Saturation / Lightness	Measure the average hue / saturation and lightness to distinguish vivid images from desaturated ones.	jit.rgb2hsl
Edge Detection	Identify boundaries, contours and transitions between light and dark regions.	jit.sobel jit.fx.sobel
Edge Density	Measure how much of an image is occupied by edges. Useful for estimating visual complexity.	jit.sobel jit.histogram
Screen Zones (Using screen position)	Section an image into quadrants, corners, halves, or strips	jit.scissors jit.glue